

SMIDOVICH, Russian Federation

WMO: 317250

Lat: 48.617N

Lon: 133.833E

Elev: 52

StdP: 100.70

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-30.6	-35.1	0.1	-32.0	-33.0	0.2	-29.9	8.1	-16.2	6.4	-15.2	1.0	230	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.8	22.9	28.2	21.8	26.7	20.9	23.9	28.0	23.0	26.8	21.9	25.4	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.5	26.3	21.6	16.4	25.3	20.7	15.5	24.2	72.4	27.8	68.6	26.9	64.8	25.5	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.2	5.3	-35.1	32.6	2.7	1.6	-37.1	33.8	-38.7	34.8	-40.2	35.7	-42.2	36.9		
			-35.2	25.7	2.7	1.5	-37.2	26.8	-38.7	27.6	-40.2	28.4	-42.1	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-21.1	-16.9	-7.4	4.4	12.5	18.1	21.5	19.8	13.5	4.5	-8.0	-19.3
	DBStd	15.77	5.11	4.80	5.90	4.21	3.85	3.26	2.54	2.77	3.64	4.24	6.29	5.29
	HDD10.0	4088	964	754	538	176	19	0	0	0	11	177	541	908
	HDD18.3	6208	1223	988	797	419	183	43	4	17	147	429	791	1166
	CDD10.0	1127	0	0	0	7	98	242	356	303	116	6	0	0
	CDD18.3	206	0	0	0	0	3	36	102	62	3	0	0	0
	CDH23.3	1516	0	0	0	4	60	323	725	375	28	1	0	0
	CDH26.7	342	0	0	0	1	7	79	194	59	2	0	0	0
Wind	WSAvg	2.2	1.8	1.9	2.4	2.7	2.6	2.2	1.9	1.9	2.1	2.4	2.4	2.0
Precipitation	PrecAvg	667	10	11	23	37	66	79	146	130	85	42	22	16
	PrecMax	935	30	41	59	106	139	205	359	246	192	112	75	68
	PrecMin	462	0	0	0	3	12	9	24	32	16	7	0	2
	PrecStd	112	8	11	17	24	34	50	74	50	43	29	18	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.9	0.1	9.9	22.6	27.6	31.5	32.6	30.1	26.0	20.6	8.3	-1.8
		MCWB	-6.8	-2.7	3.8	12.6	17.4	22.3	23.8	24.2	19.8	12.6	4.4	-3.4
	2%	DB	-7.9	-3.9	5.9	17.9	25.1	28.7	30.5	28.3	23.9	16.6	5.1	-6.6
		MCWB	-9.1	-5.9	1.6	10.4	15.3	21.1	23.4	22.6	18.0	10.2	2.2	-7.7
	5%	DB	-10.9	-6.4	3.4	14.6	22.7	26.9	28.8	26.9	22.1	13.9	2.6	-9.3
		MCWB	-11.9	-7.9	0.0	7.9	14.2	19.8	22.6	21.5	16.9	8.4	0.3	-10.3
	10%	DB	-13.3	-8.9	1.4	11.7	20.4	24.9	27.1	25.4	20.3	11.5	0.6	-11.4
		MCWB	-14.1	-10.0	-1.3	6.1	13.2	18.9	21.8	20.6	16.0	7.0	-1.1	-12.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.3	-3.1	4.3	13.3	18.9	24.1	25.3	25.2	20.9	13.6	4.8	-3.2
		MCDB	-5.2	0.0	9.1	20.7	26.3	29.4	30.2	29.0	24.4	18.5	7.3	-1.6
	2%	WB	-9.1	-5.7	2.0	10.6	16.6	22.0	24.2	23.8	19.3	11.2	2.4	-7.5
		MCDB	-7.9	-3.8	5.4	16.8	22.8	27.3	28.8	26.8	22.7	15.2	4.6	-6.5
	5%	WB	-11.9	-7.9	0.3	8.6	15.1	20.7	23.3	22.6	17.9	9.3	0.8	-10.1
		MCDB	-11.1	-6.5	2.9	13.8	20.9	25.6	27.5	25.6	20.8	12.7	2.3	-9.3
	10%	WB	-14.1	-10.0	-1.2	6.7	13.8	19.5	22.4	21.6	16.6	7.6	-0.9	-12.2
		MCDB	-13.3	-8.9	1.3	11.0	18.9	23.9	26.2	24.3	19.5	10.9	0.8	-11.4
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.8	11.0	10.5	11.1	10.3	9.2	9.2	11.1	11.2	8.9	9.2
		MCDBR	11.1	12.6	11.4	15.9	15.5	13.9	12.1	11.3	13.7	15.3	10.2	9.4
		MCWBR	10.1	10.7	7.7	9.1	7.5	7.0	6.0	6.1	7.8	9.0	7.4	8.5
	5% WB	MCDBR	11.0	12.4	10.8	14.7	13.2	12.1	10.6	9.6	11.1	13.7	9.8	9.2
		MCWBR	10.0	10.6	7.6	8.8	7.1	6.6	5.8	5.7	7.1	8.8	7.4	8.3
Clear-Sky Solar Irradiance	taub	0.284	0.326	0.401	0.543	0.567	0.520	0.507	0.448	0.413	0.438	0.362	0.297	
	taud	2.157	2.010	1.876	1.729	1.761	1.968	2.050	2.216	2.218	1.976	2.073	2.126	
	Ebn at Noon	758	795	786	708	714	756	760	792	779	652	633	676	
	Edh at Noon	82	121	165	213	214	176	160	130	117	126	90	74	
All-Sky Solar Radiation	RadAvg	1.87	3.12	4.55	4.82	5.27	5.60	5.17	4.68	3.87	2.63	1.73	1.45	
	RadStd	0.06	0.09	0.32	0.50	0.46	0.75	0.49	0.33	0.35	0.20	0.12	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page